

Questions

Part - 1

1. Identify the logo of following chatbot developed by OPEN AI:



- (a) GEMINI (b) CHATGPT
(c) BING AI (d) CLAUDE
2. What role do data science engineers play in the synergy between AI and data science engineering?
- (a) Handling electronic health records in healthcare
(b) Optimizing investment portfolios in finance
(c) Evaluating AI models using statistical methods
(d) Developing AI algorithms and models
3. Arrange the following milestones in the order of their occurrence in the field of Artificial Intelligence (AI):
1. Introduction of the term "Artificial Intelligence" by John McCarthy.
 2. IBM's Deep Blue defeats Garry Kasparov in a chess match.
 3. Development of the first neural network model.
- (a) 1 > 3 > 2 (b) 3 > 2 > 1
(c) 2 > 3 > 1 (d) 2 > 1 > 3
4. Match the following AI terms with their definitions:

Column A	Column B
1. Machine Learning	A. A branch of AI that deals with algorithms that can learn from and make predictions on data.
2. Neural Networks	B. A type of AI model inspired by the human brain's neural structure, used for pattern recognition and classification in data.
3. Natural Language Processing	C. A technique in AI that enables machines to mimic cognitive functions such as learning and problem-solving.

- (a) 1-C, 2-B, 3-A (b) 1-A, 2-B, 3-C
(c) 1-C, 2-A, 3-B (d) 1-A, 2-C, 3-B
5. A healthcare startup is developing an AI-powered diagnostic tool using neural networks to improve accuracy in detecting rare diseases.

Which aspect of neural networks is crucial for achieving this goal?

- (a) Loss functions
- (b) Activation functions
- (c) Gradient descent optimization
- (d) Regularization techniques

6. How can understanding the business processes in a given domain help optimize operations?

- (a) Enhancing decision-making
- (b) Strengthening manual errors
- (c) Enhancing fictional scenarios
- (d) Promoting irrelevant details

7. A company wants to develop an AI model to improve its sales forecasting. Which of the following must be enhanced to achieve this?

- (a) 6-month forecast
- (b) Dashboard reports
- (c) Sales managers
- (d) Model & report

8. A business wishes to improve its customer service procedure by utilizing AI. Put the steps in the proper order:

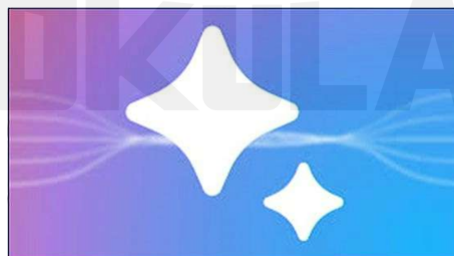
1. Put AI technologies into practice
2. Determine the main issues that customers face
3. Use consumer interaction data to train the AI model.

- (a) 3 → 1 → 2
- (b) 2 → 3 → 1
- (c) 1 → 2 → 3
- (d) 2 → 1 → 3

9. What technology trend in healthcare is described as providing near real-time replicas of lifelong data records for individuals?

- (a) Digital Twins
- (b) Robots in Healthcare
- (c) AI and Machine Learning
- (d) 3D Printing

10. Identify the following chat bot:



- (a) CLAUDE
- (b) GEMINI
- (c) BING AI
- (d) CHATGPT

11. A company wants to use AI to personalize its website content for individual users based on their browsing history. Which is the most important technical requirement for this project?

- (a) Increased staff for customer service
- (b) High-quality website design
- (c) Low operational cost
- (d) AI model for content recommendation

12. Identify the tool that can be used for creating business process models, organizational charts and other visual representations that help clarify business processes and technical requirements.



- (a) Xbox
- (b) MS Word
- (c) Notepad
- (d) Microsoft Visio

Part - 2

1. Arrange the steps in the correct order to improve customer service using AI:
 - (1) Analyze customer queries
 - (2) Implement AI chatbots
 - (3) Deploy AI solution
 - (a) $2 \rightarrow 3 \rightarrow 1$
 - (b) $3 \rightarrow 1 \rightarrow 2$
 - (c) $2 \rightarrow 1 \rightarrow 3$
 - (d) $1 \rightarrow 2 \rightarrow 3$
2. What historical event paved the way for the ability to build skyscrapers and mass-produced goods like cars?
 - (a) The Renaissance
 - (b) The French Revolution
 - (c) The Industrial Revolution
 - (d) The Enlightenment
3. A company is developing an AI assistant that can access and analyze user's personal across multiple platforms to provide personalized recommendations. Which of the following is the most significant ethical concern in this scenario?
 - (a) Transparency of AI decision-making processes
 - (b) Impact on human jobs and employment
 - (c) User privacy and data protection
 - (d) Potential bias AI algorithms
4. Which of the following AI frameworks is being determined in the below image?



- (a) Deep learning IO
 - (b) Neural networks
 - (c) Pytorch
 - (d) Python
5. A company decides to use Google Gemini AI to analyze sensitive customer data for personalized marketing. Which of the following is the most pressing concern in this scenario?

- (a) Customer privacy and data protection
- (b) Natural language processing capabilities
- (c) Integration with existing cloud infrastructure
- (d) Competition from other AI platforms

6. Which of the following cloud's AI models is demonstrated in the image below?



- (a) AWS
 - (b) Dataiku
 - (c) Azure
 - (d) Google
7. An AI-powered medical imaging system misdiagnoses a patient's condition, leading to inappropriate treatment. Which of the following is the most critical ethical consideration in this scenario?
- (a) Potential job displacement of radiologists
 - (b) Algorithm accuracy and reliability
 - (c) Transparency and accountability of AI decision-making
 - (d) Patient privacy and data security
8. A manufacturing company uses AI to optimize production schedules. Which are does AI improve?
- (a) Worker safety
 - (b) Production efficiency
 - (c) Product quality
 - (d) Facility size
9. A retail company uses AI to predict demand and manage inventory. Which are will AI improve?
- (a) Stock levels
 - (b) Warehouse construction
 - (c) Customer feedback
 - (d) Employee training
10. Which of the best is illustrated in the image below?



- (a) AI Process management
 - (b) AI Applications
 - (c) AI Future scope
 - (d) AI life cycle
11. Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) are often mentioned interchangeably, but they represent distinct aspects of technology. What distinguishes Deep Learning (DL) from Artificial Intelligence (AI) and Machine Learning (ML)?

- (a) DL involves algorithms learning from data without explicit programming.
- (b) DL focuses on achieving AI through algorithms and data training.
- (c) DL mimics brain functions with layered neural networks for pattern recognition.
- (d) DL encompasses tasks like language understanding and problem-solving.

12. Match the following AI tools with their relevant use cases in healthcare:

AI Tool	Use Case
A. Neural Network	1. Predicting Patient Outcomes Based on Data
B. Decision Trees	2. Detecting Tumors in medical images
C. Natural Languages Processing	3. Understanding and interpreting doctor's notes

- (a) 1-A, 2-C, 3-B
- (b) 1-B, 2-A, 3-C
- (c) 1-B, 2-C, 3-A
- (d) 1-C, 2-B, 3-A

13. Identify the following coding language which is used in ML:



- (a) JAVA
- (b) PYTHON
- (c) C++
- (d) C

14. What analogy does Tomaso Poggio use to describe the periodicity of scientific ideas like neural networks?

- (a) Waves in the ocean
- (b) Epidemics of viruses
- (c) Cycles of the moon
- (d) Growth of plants

15. Arrange the following steps in the correct order for developing a Machine Learning model:

1. Model Evaluation
2. Data Preprocessing
3. Model Training

- (a) 2 > 3 > 1
- (b) 2 > 1 > 3
- (c) 1 > 3 > 2
- (d) 3 > 2 > 1

16. Arrange the following steps in correct order to describe the training process of a neural network:

1. Activation Function Application
2. Backpropagation
3. Forward Propagation

(a) $2 > 1 > 3$

(b) $2 > 3 > 1$

(c) $1 > 3 > 2$

(d) $3 > 2 > 1$

17. How can AI improve a recommendation system?

(a) Use collaborative filtering

(b) Use decision trees

(c) Use clustering

(d) Use linear regression

18. Which AI method helps in predicting future sales based on past data?

(a) Clustering

(b) Data visualization

(c) Regression analysis

(d) Decision trees

Part - 3

1. Which of the following tools is best suited for creating and presenting AI-generated reports in a professional format?

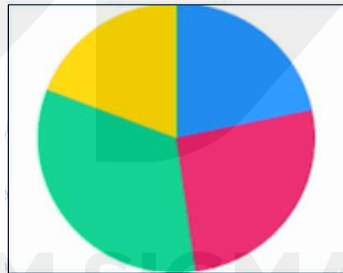
(a) PowerPoint

(b) Excel

(c) Microsoft Word

(d) Gemini

2. Which type of visualization chart is represented in the image given below?



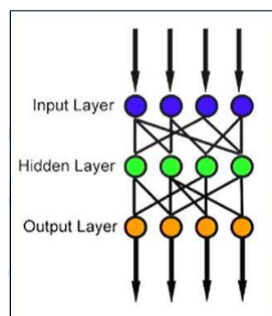
(a) Line chart

(b) Scatter plot

(c) Pie chart

(d) Doughnut chart

3. What type of AI model architecture is depicted in the image?



(a) Recurrent Neural Network (RNN)

(b) Reinforcement Learning Model

(c) Feedforward Neural Network

(d) Convolutional Neural Network (CNN)

4. What technological advancement contributed significantly to the resurgence of neural networks in recent years?
- (a) Support vector machines (SVMs) (b) Quantum computing
(c) Analog computing (d) Graphics processing units (GPUs)
5. You need to visualize the sales growth trend over the last 12 months. Which type of chart would best display this data?
- (a) Line chart (b) Bar chart
(c) Pie chart (d) Heat map
6. Match the tools in Column A with their primary uses in Column B.

Column A	Column B
1. Microsoft Word	A. Creating visual presentations
2. Google Sheets	B. Organizing and analyzing data
3. Microsoft PowerPoint	C. Writing and formatting text documents

- (a) 1-B, 2-C, 3-A (b) 1-C, 2-B, 3-A
(c) 1-A, 2-C, 3-B (d) 1-B, 2-A, 3-C
7. You have analyzed the number of hospital visits for different diseases over the past year. Which of the following visualizations would be most suitable for representing this data?
- (a) Pie chart (b) Line chart
(c) Bar chart (d) Heat map

M SIGMA
GOKULAM